

ONLINE SHOPPING MANAGEMENT SYSTEM

Project Roadmap & Documentation

Duration: 11 Weeks **Type:** Academic Project **Version:** 1.0

Technologies: PHP / Python / MySQL · Modules: Auth, Products, Cart, Orders, Payments, Admin

Project Overview

The **Online Shopping Management System** is a full-stack web application that enables customers to browse products, manage a shopping cart, place orders, and track payments, while providing administrators with a comprehensive dashboard for managing users, inventory, and reporting. The system is developed iteratively across **11 weeks**, following structured software engineering principles.

Aspect	Detail
Problem Statement	Existing manual shopping processes are slow and error-prone; the system automates order, product, and payment management.
Target Users	Customers (browse & buy), Admins (manage & report)
Core Modules	Authentication, Products, Categories, Cart, Orders, Payments, Admin Dashboard
Methodology	Incremental / Modular development with weekly deliverables
Repository	GitHub (version-controlled, branching per module)

Week-by-Week Project Roadmap

Week 1 — Project Planning & Requirement Analysis

Tasks:

- Project topic selection and problem scoping
- Problem statement writing
- Requirement gathering (stakeholder analysis)
- Define functional & non-functional requirements
- Create Use Case Diagram (actors & interactions)
- Prepare project roadmap documentation
- Create GitHub repository (branching strategy)

Project Proposal
Requirement Specification Document
Use Case Diagram
GitHub Repository (initialized)

Week 2 — System Design**Tasks:**

- Database schema planning
- ER Diagram creation (entities, relationships, cardinalities)
- System architecture design (MVC / layered)
- UI wireframe design for key screens
- Define API endpoints (if applicable)

ER Diagram
Database Schema (SQL DDL)
UI Wireframes
Architecture Diagram

Week 3 — Database Development**Tasks:**

- Create MySQL database
- Create tables: `Users`, `Products`, `Orders`, `Categories`, `Payments`, `Cart`, `Sessions`, `Order_Items`
- Define PKs, FKs, constraints, indexes
- Insert representative sample/seed data

SQL Scripts (DDL + DML)
Database Setup Documentation
Seed Data File

Week 4 — User Authentication Module**Tasks:**

- User registration (form validation, duplicate check)
- Login system (credential verification)
- Password security (bcrypt hashing, salting)
- Session management (tokens, expiry, logout)
- Role-based access control (customer vs admin)

Registration & Login Features
Auth Middleware
Password Reset Flow

Week 5 — Product Management Module**Tasks:**

- Add products (name, price, stock, image upload)
- Edit product details
- Soft delete products
- Product listing page (pagination)
- Stock quantity management

Product Management System
CRUD Interface
Image Upload Handler

Week 6 — Category & Search Module**Tasks:**

- Product category management (hierarchical)
- Full-text search functionality
- Product filtering (price range, rating, brand)
- Sorting (price, newest, popularity)

Search & Filter Features
Category Browser
Filter UI Components

Week 7 — Shopping Cart Module**Tasks:**

- Add item to cart (with stock validation)
- Remove item from cart
- Update item quantity
- Cart summary (subtotal, tax, total)
- Persistent cart (database-backed)

Shopping Cart System
Cart Summary Page
Stock Validation Logic

Week 8 — Order Management Module**Tasks:**

- Place order from cart (checkout flow)
- Order history view for users
- Order status tracking (Pending → Shipped → Delivered)
- Automatic stock deduction on order
- Order confirmation notification

Order Processing System

Order History Page

Status Tracking UI

Week 9 — Payment Module**Tasks:**

- Payment interface (checkout page)
- Cash on Delivery (COD) implementation
- Online payment gateway integration (optional)
- Payment record storage
- Receipt / invoice generation

Payment Management System

COD Module

Payment Records & Receipts

Week 10 — Admin Dashboard**Tasks:**

- User management (view, activate, deactivate)
- Product management from admin panel
- Order management (update statuses)
- Sales report generation
- Dashboard analytics (charts, KPIs)

Admin Panel

Sales Reports

Analytics Dashboard

Week 11 — Testing & Final Documentation**Tasks:**

- System testing (unit, integration, UAT)
- Bug identification and fixing
- Performance checking & optimization
- Final report writing (IEEE / university format)
- Presentation preparation (slides)
- Final submission packaging

Final Report

Presentation Slides

Complete Project Submission

Test Case Documents

Project Timeline (Gantt Overview)

Module / Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11
Planning & Analysis	•										
System Design		•									
Database Dev			•								
Authentication				•							
Product Management					•						
Category & Search						•					
Shopping Cart							•				
Order Management								•			
Payment Module									•		
Admin Dashboard										•	
Testing & Docs											•

Functional Requirements

ID	Module	Requirement
FR01	User Auth	Users must be able to register with unique email and password.
FR02	User Auth	Users must be able to log in and receive a session token.
FR03	User Auth	Passwords must be stored as bcrypt hashes (min. 60-char digest).
FR04	Products	Admins can add, edit, and soft-delete products.
FR05	Products	Product listing must support pagination (20 items per page).
FR06	Categories	Products must be assigned to at least one category.
FR07	Search	Users can search products by keyword (name, brand, description).
FR08	Search	Filter by price range, rating, and category must be available.
FR09	Cart	Authenticated users can add/remove/update cart items.

ID	Module	Requirement
FR10	Cart	Cart must validate available stock before checkout.
FR11	Orders	Users can place orders from their cart (checkout).
FR12	Orders	Order status progresses: Pending → Processing → Shipped → Delivered.
FR13	Payments	System supports Cash on Delivery as minimum payment method.
FR14	Payments	Each order must produce a payment record and receipt.
FR15	Admin	Admin can view all users, orders, and generate sales reports.

Non-Functional Requirements

Category	Requirement
Performance	Page load time ≤ 3 seconds for product listings; DB queries optimized with indexes.
Security	SQL injection prevention (prepared statements), XSS protection, HTTPS enforced.
Usability	Responsive UI (mobile, tablet, desktop); WCAG 2.1 AA accessibility minimum.
Reliability	System uptime $\geq 99\%$ during testing phase; graceful error handling.
Scalability	Database normalized to 3NF; architecture supports future microservices split.
Maintainability	Code follows MVC pattern; inline comments; minimum 80% test coverage target.

Database Entities Summary

Table	Primary Key	Description
Users	user_id (INT)	Stores customer and admin accounts, credentials, roles.
Categories	category_id (INT)	Hierarchical product categories with parent reference.
Products	product_id (INT)	Product catalog with pricing, stock, images, ratings.
Cart	cart_id (INT)	Per-user cart items linking user and product records.
Orders	order_id (INT)	Order headers with user, total amount, status, address.
Order_Items	item_id (INT)	Line items per order: product, quantity, unit price.
Payments	payment_id (INT)	Payment method, amount, status, transaction reference.
Sessions	session_id (INT)	Auth session tokens, IP, creation time, expiry.

Total Project Deliverables Checklist

Planning & Design Project Proposal Requirement Specification Use Case Diagram ER Diagram Database Schema UI Wireframes	Development Modules User Authentication Product Management Category & Search Shopping Cart Order Management Payment Module Admin Dashboard
Database SQL DDL Scripts Sample Data (DML) DB Setup Documentation	Final Submission Final Technical Report Presentation Slides Test Case Documents Complete Source Code (GitHub)